

Section 3 – Data and Empirical Strategy

Purpose of this report

This report turns the high-level outline for **Section 3 (Data and Empirical Strategy)** of your Econ H191 final paper into a detailed plan. For each subsection, it:

1. Clarifies what the subsection should do in the final paper.
2. Specifies which parts of the **thesis proposal** (and related context PDFs) can be reused or adapted.
3. Flags where **new writing** is required and what that writing should emphasize.
4. Keeps the structure consistent with the **Thesis Design Redesign Report** and the **Mega Context & Handoff** documents, so that Section 3 meshes cleanly with Sections 1–2 (Introduction, Literature Review) and Section 4 (Results).

The goal is that you can now: - Lift/adapt text from the proposal where appropriate. - Know exactly where to update language for the new LP + K/L design and mediation model. - Avoid redundancy with Sections 1–2.

I'll follow the section numbering you proposed:

1. Data and Empirical Strategy

- 3.1 Overview 3.2 Data sources 3.3 Construction of key variables 3.4 Econometric framework
3.5 Identification assumptions and threats 3.6 Robustness and sensitivity analyses
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3.1 Overview

Function in the final paper

A short orientation section that: - Summarizes, in 1–2 paragraphs, **what data** you use (port/terminal–month panel, throughput, labor, capital, controls). - States the **three main outcomes**/objects of interest: - $\ln(LP)$ = log output per worker. - $\ln(K/L)$ = log capital per worker. - The **elasticity of LP with respect to K/L**. - Signals the **two-level econometric approach**: - Event-study DiD for LP and K/L (Models 1A and 1B). - Panel FE regression for LP on K/L (Model 2), plus the mediation-style decomposition.

This should read like a roadmap for Section 3, not a repetition of the Introduction.

What to reuse from the proposal

- Use the general style of the **current proposal's Section 5 ("Data") opening paragraphs** and **Section 6.1 ("Econometric Design") opening paragraph**, but shorten and reframe them:
- Instead of treating only LP as the outcome, explicitly mention **both LP and K/L**.
- Remove discussion of instruments and engineering milestones as IV; those are no longer framed as instruments but as inputs to K construction.

What requires new writing

- A compact paragraph explicitly stating that:
 - You work with a **terminal-month panel** (with some port-level pre-split periods).
 - You use **administrative throughput data, labor data** (currently proxy, to be replaced by true hours), and a newly constructed **monthly capital series** for Haifa legacy.
 - The econometric analysis proceeds in two stages:
 - Step 1: Reforms → LP and K/L using **not-yet-treated event-studies**.
 - Step 2: LP ↔ K/L relationship using a **fixed-effects panel regression**.
 - A sentence or two explaining that the **detailed construction** of LP and K/L is in Sections 3.2–3.3, and the **formal specifications** are in Section 3.4.
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3.2 Data sources

Function in the final paper

This subsection should give the reader a clear, concise inventory of all **data sources**, at a level slightly above the code/QA detail. It should: - Make clear what is **observed** versus **constructed**. - Separate **throughput/output, labor, capital**, and **controls**. - Flag data limitations (coverage, frequency) that matter for the econometrics.

Proposed substructure

- 3.2.1 Throughput and operational data
- 3.2.2 Labor data
- 3.2.3 Capital and project-level data
- 3.2.4 Time controls and other covariates

3.2.1 Throughput and operational data

Goal: Describe the port/terminal-month throughput panel and any auxiliary operational measures.

Reuse from proposal: - The description in the **proposal's Data section** where you explain the TEU and tonnage data: source (Ports Authority / IPC / shipping statistics), level of disaggregation (port vs terminal, pre-/post-split), period coverage (roughly 2016–2024 or your chosen window), and how ship calls/ship waiting times (if used) are defined.

Updates / new writing: - Clarify that **TEU and tonnage series are now stable and treated as “final” inputs**. - Explicitly connect this to the **LP and L_{proxy} pipelines**: - Throughput Y enters as the numerator of LP. - The same Y is used in calibrating the labor proxy (TEU/hour KPIs) and in aligning K timing (e.g., post-project throughput changes). - Briefly mention the **port vs. terminal split** and how you handle pre-split periods (e.g., Haifa pre-Bayport, Ashdod pre-HCT) by working at the port level before the terminals are separately observed.

3.2.2 Labor data

Goal: Explain clearly what labor data you currently use, what you expect to receive, and why the **design is robust to updating L** later.

Reuse from proposal: - Parts of the proposal's discussion of the **TEU/hour KPI-based labor proxy**: how KPIs are recorded, how you invert them into labor-hours, how you aggregate to terminal-month, and the main QA checks.

Updates / new writing: - Explicitly flag that: - The current LP estimates are based on a **proxy L** inferred from TEU/hour KPIs. - You expect to replace this with **actual monthly labor-hours** once data arrive from IPC/HPC/APC. - The **econometric structure does not change** when you swap in true L; only measurement of LP and K/L improves. - Add 1–2 sentences on why proxy L is **problematic for mediation/IV** (L appears in both LP and K/L, and TEU/hour-based L may respond mechanically to throughput changes), motivating the emphasis on true L in the final version. - Clarify what you will report in the thesis: - LP results based on **final L**, with the proxy only used as a robustness or design check.

3.2.3 Capital and project-level data

Goal: Describe the raw inputs used to build the capital series, without yet going into full PIM detail.

Reuse from proposal and methods docs: - Proposal's brief description of capital measurement (if any) can be used as a starting point but needs to be fully updated. - Heavily adapt from: - **"Haifa Legacy K-series Construction Plan"** and - **"HPC Track-B Raw Extraction & K-series Construction – Methods Report (final)"** where you list the financial statement sources, Note-8 PPE flows, and the big-projects table.

Updates / new writing: - Make clear that the **capital series is currently constructed only for Haifa's legacy container operations** (and potentially extended later), and that it relies on: - Annual **Haifa Port Company financial statements** (Note-8 PPE, cash-flow statements). - A **national capital deflator** (GFCF index) to move from nominal to real. - A **project-level dataset** of big investments and disposals (STS cranes, berth extensions, marine department sale, etc.) with commissioning dates. - Explain in non-technical prose how Path B works at a high level: - Annual PIM anchored in financials. - Monthly interpolation of background investment. - Lumpy shocks at commissioning dates of major projects. - Emphasize that the detailed algorithm is deferred to **an appendix** (with references to the methods memo), so that Section 3 remains readable.

3.2.4 Time controls and other covariates

Goal: Briefly list and motivate other variables that enter X_{it} or Z_{it} in the regressions.

Reuse from proposal: - Any existing mention of **COVID, war/security events, strikes, regulatory changes**, or macro shocks that might affect throughput and LP.

Updates / new writing: - Provide a concise table or bullet list in text: - COVID dummies / lockdown periods. - War/security episodes that materially disrupted port activity. - National holidays / seasonal controls if used. - Any macro controls (e.g., world trade proxies) if you decide to include them. - Clarify whether these enter

as **calendarmonth fixed effects** (absorbing a lot of aggregate variation) versus specific dummies for particularly salient shocks.

3.3 Construction of key variables

Function in the final paper

This subsection is where you **formally define** the main variables used in the econometrics. It should: - Give clear formulas for LP, K, and K/L. - Explain how you go from raw data to the panel used in estimation (stacking, quarterization if relevant, etc.). - Clarify how legacy periods and terminal splits are handled.

Proposed substructure

3.3.1 Output and measurement of labor productivity (LP)

3.3.2 Labor input: from proxy L to true L

3.3.3 Capital stock and capital-to-labor ratios (K/L)

3.3.4 Final analysis panels and sample restrictions

3.3.1 Output and measurement of labor productivity (LP)

Goal: Define LP precisely and tie it to your throughput data.

Reuse from proposal: - The proposal's definition of **LP = Y/L** at the terminal-month level, where Y is constructed as a **mix-weighted aggregation of TEU and tonnage**, and L is labor-hours. - The description of how you handle **mixed units** (TEU vs tons) and the construction of a composite output measure with weights derived from historical tons/TEU ratios (winsorized), if this is part of your current LP pipeline.

Updates / new writing: - Clarify that you work with **log LP** as the main outcome: $\ln(LP_{it}) = \ln(Y_{it}) - \ln(L_{it})$. - Make explicit the **unit of observation** and time dimension (terminal-month, with some port-month pre-split) and the calendar coverage used for estimation (e.g., 2016–2024 excluding extreme COVID months if you choose to drop them). - Explain briefly the **source precedence rules** used in building the canonical panel (e.g., port-level series vs terminal-level series, how you prioritize sources when multiple LP measures exist for a given period).

3.3.2 Labor input: from proxy L to true L

Goal: Go deeper into how L is constructed and how it will be updated.

Reuse from proposal: - The more technical parts of the proposal's **labor proxy construction** (L_{proxy}) can be moved here or summarized and then referenced to an appendix: - How TEU/hour KPIs are inverted to give total hours. - How KPIs are matched to terminals and months. - QA checks comparing annual implied hours to reported staffing or shift patterns.

Updates / new writing: - Add a short subsection explicitly titled **"From proxy L to true L"**, where you: - State that Method A (proxy L) is fully implemented and used for design tests. - State that Method B (true L)

will simply replace L in the LP and K/L formulas once monthly hours are obtained, without changing the econometric design. - Mention any **anticipated adjustments** when true L arrives: - E.g., differences in coverage (which workers are included), timing (when hours are posted), or scope (port-level vs terminal-level). - Clarify which version of L will be used for the **final thesis tables and figures**, and how you will treat the proxy-based results (e.g., as robustness or appendix).

3.3.3 Capital stock and capital-to-labor ratios (K/L)

Goal: Translate the Path B methodology into paper-ready definitions of K and K/L.

Reuse from methods docs: - Draw lightly (and in prose) from the **Haifa Legacy K-series Construction Plan** and **HPC Track-B Methods Report**: - PIM equation in words. - Explanation of how additions, disposals, and depreciation from financials feed into annual K. - Explanation of how big projects are treated as lumpy shocks at commissioning.

Updates / new writing: - Define the **real capital stock** K_{it} at the port/terminal-month level as: - The result of a PIM procedure applied to deflated PPE, with background investment spread across months and big projects treated as jumps. - Define **capital per worker**: - $K/L_{it} = K_{it} / L_{it}$. - $c_{it} = \ln(K/L_{it}) = \ln(K_{it}) - \ln(L_{it})$. - Explain how you deal with: - The initial capital stock (base year). - Revaluations or breaks in series. - Mapping port-level K to the relevant **legacy terminal** when the split occurs (e.g., Haifa legacy container terminal inherits a specific share of port-level capital). - Flag that **uncertainty in project cost and timing** is handled through sensible assumptions and will be explored in robustness (alternative depreciation rates, alternative allocations of project costs across terminals).

3.3.4 Final analysis panels and sample restrictions

Goal: Describe the final estimation samples and any trimming/censoring decisions.

Reuse from proposal: - Parts of the proposal that describe: - The time window used in the NYT event-studies (e.g., 24 months pre, 24 months post, or whatever you choose). - The way you restrict the sample to not-yet-treated units for each reform.

Updates / new writing: - Clearly define: - The **main analysis sample** for Models 1A and 1B (units, time range, which terminals/ports included). - Any **exclusions** (e.g., outlier months of war, extreme COVID months) and the justification. - The **sample used for Model 2** (potentially the full panel, but you might choose to restrict it to periods where K_{it} is well measured and where L is final). - If you create multiple derived panels (e.g., port-quarter aggregates, stacked event-study datasets), describe them briefly and map them to specific models.

3.4 Econometric framework

Function in the final paper

This subsection is the formal backbone of your empirical strategy. It should: - Present the **event-study DiD specifications** for LP and K/L (Models 1A and 1B). - Present the **panel regression of LP on K/L** (Model 2). -

Explain the role of fixed effects, clustering, and any weighting. - Define the **mediated / accounted-for component** of LP changes via K/L.

Proposed substructure

3.4.1 Event-study DiD for labor productivity (Model 1A)

3.4.2 Event-study DiD for capital per worker (Model 1B)

3.4.3 Elasticity of LP with respect to K/L (Model 2)

3.4.4 Mediation-style decomposition

3.4.1 Event-study DiD for labor productivity (Model 1A)

Goal: Adapt the proposal's current econometric design to the new twin-outcomes framework.

Reuse from proposal: - The existing **Section 6.1 Econometric Design** is an excellent starting point: - Definition of the **not-yet-treated (NYT)** design. - Event-time variable $M_{\{i,t\}}$ and omitted bin $m = -1$. - The baseline regression for $\ln(LP_{\{i,t\}})$ with terminal and calendar-month fixed effects. - The narrative explanation of Table 2 (treatment vs control windows).

Updates / new writing: - Rewrite the model equation in a notation consistent with the redesigned thesis (e.g., β^{LP}_m for LP effects, with reform-specific superscripts r if you keep separate clocks). - Make explicit that: - You estimate **separate event-studies by reform** (Bayport entry, HCT entry, Haifa privatization). - Optionally, you may also estimate a **stacked event-study** that pools events of the same type. - Include a brief statement about **clustering standard errors** (e.g., by terminal, or by port×reform) and any decisions about weighting (e.g., unweighted vs TEU-weighted).

3.4.2 Event-study DiD for capital per worker (Model 1B)

Goal: Show that you apply the same event-study logic to $\ln(K/L_{\{i,t\}})$.

Reuse from proposal: - Reuse almost all of the narrative surrounding Model 1, but replace the outcome with $\ln(K/L_{\{i,t\}})$ and update language to stress that you are interested in the **capital-deepening response** to reforms.

Updates / new writing: - Define explicitly: - $c_{\{i,t\}} = \ln(K/L_{\{i,t\}})$, and write the regression: - $c_{\{i,t\}} = \sum_{\{m \in M_r, m \neq -1\}} \beta^{K/L}_m 1\{M_r = m\} + \gamma_i + \delta_t + \epsilon_{\{i,t\}}^{K/L}$ the **LP and K/L event paths side-by-side** for each reform and summarize them in tables using **binned averages** over post-reform horizons.

3.4.3 Elasticity of LP with respect to K/L (Model 2)

Goal: Introduce the panel regression that estimates the elasticity of LP with respect to K/L, conditional on fixed effects and controls.

Reuse from design documents: - The **Thesis Design Redesign Report** and **Mega Context & Handoff** both already define Model 2 conceptually. You can translate that language into a compact formal subsection.

Updates / new writing: - Present the baseline specification: $\ln(LP_{i,t}) = \alpha_i + \tau_t + \eta \ln(K/L_{i,t}) + Z_{i,t}'\lambda + u_{i,t}$, where α_i are unit fixed effects, τ_t are time fixed effects, and $Z_{i,t}$ includes reform dummies or event-time indicators. - Explain in words that: - η is interpreted as the **elasticity of output per worker with respect to capital per worker**, conditional on the fixed effects and $Z_{i,t}$. - This is analogous to the capital share in a Cobb–Douglas production function. - Discuss choices about: - Whether $\ln(K/L)$ enters contemporaneously or with a lag. - Whether you estimate this on the **full panel** or only on a **restricted sample** (e.g., excluding extreme disruptions).

3.4.4 Mediation-style decomposition

Goal: Connect Models 1A, 1B, and 2 to produce the decomposition of LP responses into components associated with K/L versus residual channels.

Reuse from design documents: - You already have a clear verbal and symbolic description of the mediation-style decomposition in the **Thesis Design Redesign Report**; this can be adapted directly into the paper.

Updates / new writing: - Define, for each reform r and horizon $[a,b]$: - The **total LP effect** $TE^{\{Y,r\}}_{[a,b]}$ as the average of $\beta^{\{LP,r\}}_m$ over $m \in [a,b]$. - The **change in K/L** $\Delta C^{\{r\}}$ as the average of $\beta^{\{K/L,r\}}_m$ over the same m . - The estimated elasticity η . - Define the **implied mediated component**: $IE^{\{r\}} \approx \eta \cdot \Delta C^{\{r\}}_{[a,b]}$. - Define the **residual component**: $DE^{\{r\}} \approx TE^{\{Y,r\}}_{[a,b]} - IE^{\{r\}}$. - Emphasize that this is **not a formal IV-based mediation analysis**, but a structured accounting exercise that quantifies how much of the LP response is consistent with the estimated η and observed changes in K/L.

3.5 Identification assumptions and threats

Function in the final paper

This subsection makes the design credible by: - Stating the key identifying assumptions behind the NYT event-studies and Model 2. - Discussing potential threats (differential trends, overlapping reforms, anticipatory effects, measurement error) and how you address them.

Proposed substructure

3.5.1 Assumptions for Models 1A and 1B

3.5.2 Assumptions and limitations for Model 2

3.5.3 Overlapping reforms and stacked event-studies

3.5.1 Assumptions for Models 1A and 1B

Reuse from proposal: - The proposal already discusses: - The **parallel trends** / no-simultaneous-shocks condition embedded in the NYT design. - Pre-trend checks in the event-study graphs. - The rationale for using not-yet-treated units as controls instead of already-treated ones.

Updates / new writing: - Expand the discussion to cover **both LP and K/L**: - The identifying assumption is that, absent the reform, treated and not-yet-treated terminals would have followed similar trends in LP and in K/L. - Address specific threats in this port setting: - Differential demand shocks across Haifa and Ashdod. - National policies that differentially favor one port. - Changes in shipping patterns unrelated to reforms. - State how you will use **pre-trend tests** and **placebo windows** to evaluate these assumptions.

3.5.2 Assumptions and limitations for Model 2

Reuse from design documents: - The conceptual discussion of Model 2 in the redesign report already emphasizes that it is **not an IV model**, but a conditional association.

Updates / new writing: - Explicitly note that: - η is estimated under the assumption that, conditional on fixed effects and $Z_{i,t}$, variation in $\ln(K/L_{i,t})$ is exogenous with respect to transitory shocks to $\ln(LP_{i,t})$, or at least sufficiently exogenous for an accounting-style exercise. - Reverse causality ($LP \leftrightarrow K/L$) is a concern; you mitigate this partially by using lags or by focusing on medium-run movements. - Be transparent that Model 2 is **interpretive**, and that you will cross-check η against **capital-income shares** from Haifa accounts.

3.5.3 Overlapping reforms and stacked event-studies

Reuse from proposal: - Any existing discussion of multiple reforms and the need to isolate each reform's effect.

Updates / new writing: - Describe how **overlapping reforms** (e.g., Bayport entry and Haifa privatization in close succession) can complicate identification. - Outline your plan to: - Use **separate event-study clocks** for each reform. - Potentially implement **stacked event-studies** that treat each reform instance as a separate event and then pool them. - Mention that you will conduct sensitivity checks with: - Different event windows (short vs long post periods). - Dropping periods where multiple reforms are simultaneously in effect.

3.6 Robustness and sensitivity analyses

Function in the final paper

This subsection previews the key robustness and sensitivity checks that you will later report in Section 4 and appendices. It reassures the reader that the main results are not driven by fragile choices.

Proposed substructure

- 3.6.1 Alternative specifications and samples
- 3.6.2 Alternative capital and labor assumptions
- 3.6.3 Placebo tests and falsification exercises

3.6.1 Alternative specifications and samples

Reuse from proposal: - Any mention of robustness checks in the current proposal (e.g., alternative event windows, inclusion/exclusion of certain terminals, different ways of grouping months).

Updates / new writing: - Explicitly list: - Alternative binning of event time (e.g., monthly vs multi-month bins, different pre/post horizons). - Alternative fixed-effects structures (e.g., adding port×quarter trends). - Alternative clustering schemes for standard errors. - Sensitivity to sample restrictions (dropping extreme disruption periods, using only legacy terminals, etc.).

3.6.2 Alternative capital and labor assumptions

Goal: Show that your results are robust to key modeling choices in K and L.

Reuse from methods docs: - The methods reports already contain suggestions for alternative depreciation rates and different allocations of project costs.

Updates / new writing: - For capital: - Vary the **depreciation rate** in the PIM (e.g., 4%, 6%, 8%). - Vary the treatment of big projects (e.g., different commissioning months within a narrow window). - Potentially try a simpler **annual K** series (no lumpy shocks) as a bounding case. - For labor: - Compare LP and K/L results using **proxy L** vs **true L** (when available). - Explore alternative ways of defining L (e.g., different allocations of port-level hours to terminals) if the raw labor data are not perfectly granular.

3.6.3 Placebo tests and falsification exercises

Goal: Enhance credibility by showing that the design does not produce spurious effects where none should exist.

Reuse from proposal: - Any existing mention of placebos (e.g., fake reform dates, non-affected terminals).

Updates / new writing: - Propose concrete tests, for example: - **Pre-reform placebo events:** assign a fake reform date 12–24 months before the true one and run the same event-study; show coefficients are near zero. - **Non-treated outcome/placebo variables:** e.g., run the reform event-study on an outcome that should not be affected by port reforms (if available). - **Cross-port placebos:** e.g., use Bayport's opening as a placebo in Ashdod for the pre-HCT period, checking that it does not generate large LP or K/L responses.

How this report should guide your drafting

1. **Start with 3.1–3.2:**
2. Write a compact 3.1 overview drawing on the proposal's Data and Econometric Design sections, but updated for LP + K/L.
3. Draft 3.2 with one paragraph per sub-subsection, using proposal text for throughput and L_{proxy}, and new prose for the capital sources.

4. Then 3.3:

5. Move the more technical LP and L constructions from the proposal into 3.3.1–3.3.2.
6. Translate the Path B capital-construction memos into 3.3.3 in a readable, high-level way, and defer details to an appendix.
7. Close with a crisp description of your final analysis samples in 3.3.4.

8. Next 3.4:

9. Recast the proposal's Section 6.1 into Models 1A and 1B, and add the short Model 2 subsection plus mediation definitions, closely following the Design Redesign Report.

10. Finally 3.5–3.6:

11. Elevate and expand the identification discussion from the proposal, now covering both LP and K/L and the mediation step.
12. Lay out robustness plans clearly so that later, when results are in, you can simply insert references to figures/tables rather than re-argue the logic.

Once you draft the text for each subsection, we can iteratively refine tone, tighten language, and check that there is no redundancy with Sections 1–2 or 4.